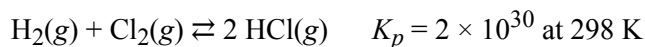
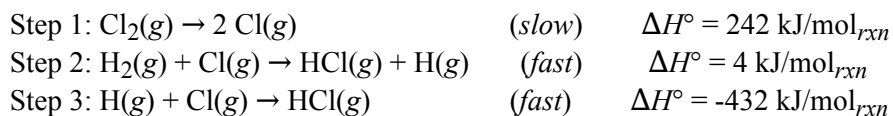


Chapter 10 Part 1: Keq



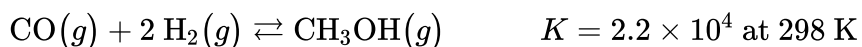
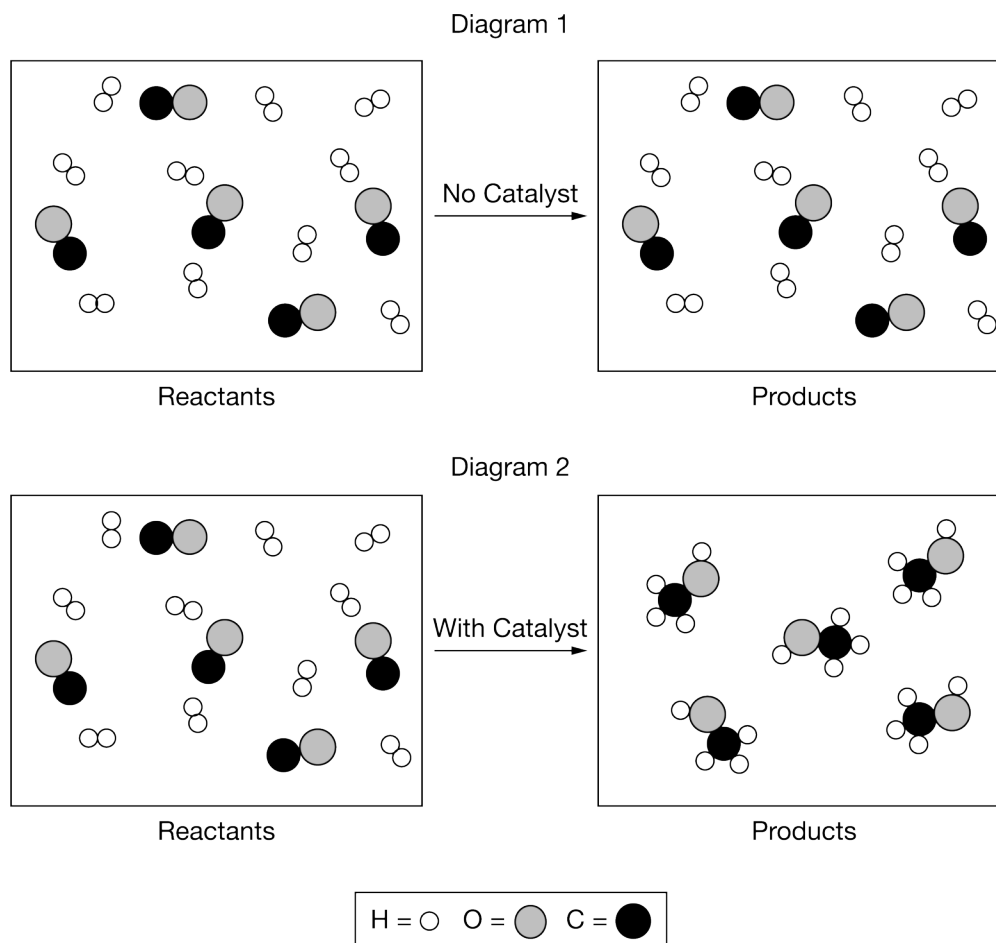
HCl(g) can be synthesized from H₂(g) and Cl₂(g) as represented above. A student studying the kinetics of the reaction proposes the following mechanism



1. Which of the following statements identifies the greatest single reason that the value of K_p for the overall reaction at 298 K has such a large magnitude?
- (A) The activation energy for step 1 of the mechanism is large and positive.
 - (B) The activation energy for step 2 of the mechanism is small and positive.
 - (C) The value of ΔS° for the overall reaction is small and positive.
 - (D) The value of ΔH° for the overall reaction is large and negative.
-

Chapter 10 Part 1: Keq

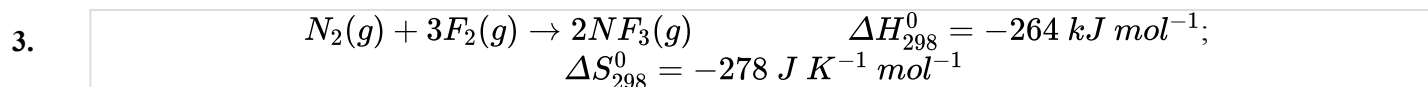
2.



A stoichiometric mixture of $\text{CO}(g)$ and $\text{H}_2(g)$ was allowed to react in two different 2.0 L rigid containers at a constant temperature of 298 K. The reaction is represented by the equation above. Diagram 1 represents the uncatalyzed reaction and diagram 2 represents the catalyzed reaction one hour after the reactants were mixed. Which of the following correctly explains the experimental results represented in the particle diagrams?

- (A) Although the reaction is thermodynamically favorable because $\Delta G^\circ < 0$ based on the value of K , only the catalyzed reaction could proceed in one hour because its reactant molecules had a higher average kinetic energy.
- (B) Although the reaction is thermodynamically favorable because $\Delta G^\circ < 0$ based on the value of K , only the catalyzed reaction could proceed in one hour because it has a lower activation-energy reaction pathway.
- (C) The reaction is not thermodynamically favorable because $\Delta G^\circ > 0$ based on the value of K , but the addition of a catalyst improved the orientation of the reactants during collisions, allowing the catalyzed reaction to proceed in one hour.
- (D) The reaction is not thermodynamically favorable because $\Delta G^\circ > 0$ based on the value of K , but the catalyzed reaction could proceed in one hour because it has a lower ΔH and a higher ΔS .

Chapter 10 Part 1: Keq



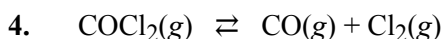
The following questions relate to the synthesis reaction represented by the chemical equation in the box above.

- Calculate the value of the standard free energy change, ΔG_{298}^0 , for the reaction.
- Determine the temperature at which the equilibrium constant, K_{eq} , for the reaction is equal to 1.00 . (Assume that ΔH^0 and ΔS^0 are independent of temperature.)
- Calculate the standard enthalpy change, ΔH^0 , that occurs when a 0.256 mol sample of $NF_3(g)$ is formed from $N_2(g)$ and $F_2(g)$ at 1.00 atm and 298 K.

The enthalpy change in a chemical reaction is the difference between energy absorbed in breaking bonds in the reactants and energy released by bond formation in the products.

- How many bonds are formed when two molecules of NF_3 are produced according to the equation in the box above?
- Use both the information in the box above and the table of average bond enthalpies below to calculate the average enthalpy of the F – F bond.

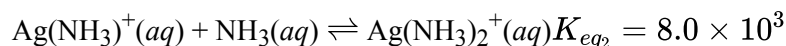
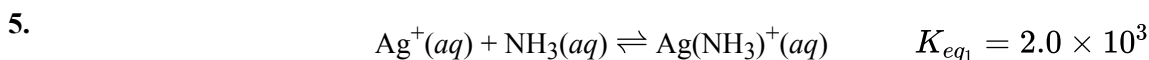
Bond	Average Bond Enthalpy (kJ mol ⁻¹)
N≡N	946
N–F	272
F–F	?



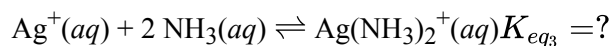
$COCl_2(g)$ decomposes according to the equation above. When pure $COCl_2(g)$ is injected into a rigid, previously evacuated flask at 690 K, the pressure in the flask is initially 1.0 atm. After the reaction reaches equilibrium at 690 K, the total pressure in the flask is 1.2 atm. What is the value of K_p for the reaction at 690 K?

- 0.040
- 0.050
- 0.80
- 1.0

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Equal volumes of 0.1 *M* AgNO₃(aq) and 2.0 *M* NH₃(aq) are mixed and the reactions represented above occur. Which Ag species will have the highest concentration in the equilibrium system shown below, and why?



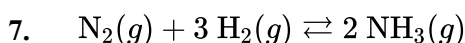
- (A) Ag⁺(aq), because $K_{eq3} = 4$
- (B) Ag⁺(aq), because $K_{eq1} < K_{eq2}$
- (C) Ag(NH₃)₂⁺(aq), because $K_{eq3} = 1.6 \times 10^7$
- (D) Ag(NH₃)₂⁺(aq), because $K_{eq1} < K_{eq2}$



Equal volumes of 0.1 *M* AgNO₃(aq) and 2.0 *M* NH₃(aq) are mixed and the reactions represented above occur. Which Ag species will have the highest concentration in the equilibrium system shown below, and why?



- (A) Ag⁺(aq), because $K_{eq3} = 4$
- (B) Ag⁺(aq), because $K_{eq1} < K_{eq2}$
- (C) Ag(NH₃)₂⁺(aq), because $K_{eq3} = 1.7 \times 10^7$
- (D) Ag(NH₃)₂⁺(aq), because $K_{eq1} < K_{eq2}$



$$K = 5.6 \times 10^5 \text{ at } 298 \text{ K}$$

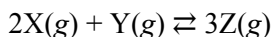
$$\Delta H^\circ_{rxn} = -91.8 \text{ kJ/mol}_{rxn}$$

The synthesis of NH₃ is represented by the equation above. Based on the equilibrium constant, *K*, and ΔH°_{rxn} given above, which of the following can best be used to justify that the reaction is thermodynamically favorable at 298 K and constant pressure?

Chapter 10 Part 1: Keq

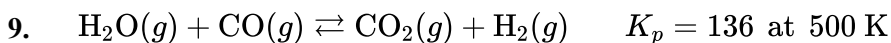
- (A) $\Delta G^\circ = -RT \ln K > 0$ because $K \gg 1$
 (B) $\Delta G^\circ = -RT \ln K < 0$ because $K \gg 1$
 (C) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ < 0$ because $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$
 (D) $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ > 0$ because $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$

8.



The reaction mixture represented above is at equilibrium at 298 K, and the molar concentrations are $[X] = 2.0 M$, $[Y] = 0.5 M$, and $[Z] = 4.0 M$. What is the value of the equilibrium constant for the reaction at 298 K?

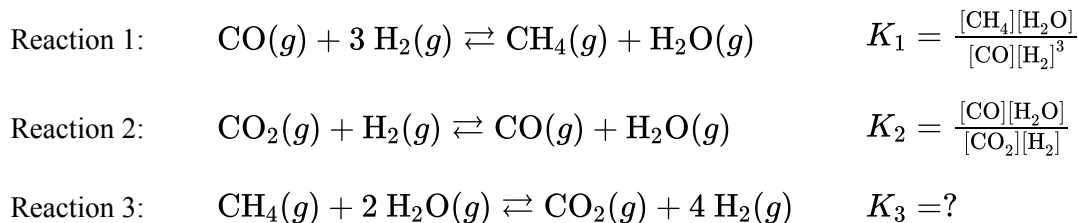
- (A) 0.50
 (B) 2.0
 (C) 4.0
 (D) 16
 (E) 32



A vessel initially contains $H_2O(g)$ at a partial pressure of 0.30 atm, $CO(g)$ at a partial pressure of 0.10 atm, $CO_2(g)$ at a partial pressure of 1.5 atm, and $H_2(g)$ at a partial pressure of 10. atm at 500 K. Which of the following occurs as the system approaches equilibrium at 500 K?

- (A) The partial pressures of $H_2O(g)$ and $CO(g)$ increase because $Q > K_p$.
 (B) The partial pressures of $H_2O(g)$ and $CO(g)$ increase because $Q < K_p$.
 (C) The partial pressures of $CO_2(g)$ and $H_2(g)$ increase because $Q > K_p$.
 (D) The partial pressures of $CO_2(g)$ and $H_2(g)$ increase because $Q < K_p$.

10.



The chemical equations and equilibrium expressions for two reactions at the same temperature are given above. Based on the information, which of the following expressions can be used to calculate the value of K_3 for reaction 3 at the same temperature?

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- (A) $K_3 = (-K_1) + (-K_2)$
- (B) $K_3 = (-K_1) - (-K_2)$
- (C) $K_3 = K_1 \times K_2$
- (D) $K_3 = \frac{1}{K_1} \times \frac{1}{K_2}$

11.



The synthesis of $\text{CH}_3\text{OH}(g)$ from $\text{CO}(g)$ and $\text{H}_2(g)$ is represented by the equation above. The value of K_c for the reaction at 483 K is 14.5.

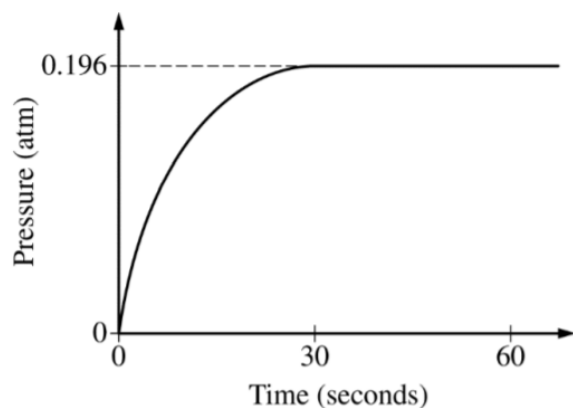
A 1.0 mol sample of $\text{CO}(g)$ and a 1.0 mol sample of $\text{H}_2(g)$ are pumped into a rigid, previously evacuated 2.0 L reaction vessel at 483 K. Which of the following is true at equilibrium?

- (A) $[\text{H}_2] = 2[\text{CO}]$
- (B) $[\text{H}_2] < [\text{CO}]$
- (C) $[\text{CO}] = [\text{CH}_3\text{OH}] < [\text{H}_2]$
- (D) $[\text{CO}] = [\text{CH}_3\text{OH}] = [\text{H}_2]$

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Substance	Lewis Diagram	Boiling Point
CH ₃ OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$	338 K
C ₂ H ₅ OH	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	351 K

Equimolar samples of CH₃OH(*l*) and C₂H₅OH(*l*) are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing CH₃OH(*l*) is shown below.

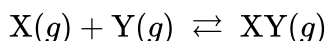


12. The value of K_p for the evaporation of CH₃OH(*l*) at 300 K is closest to
- (A) 0.04
(B) 0.2
(C) 5
(D) 30

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13.

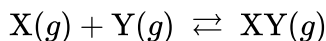
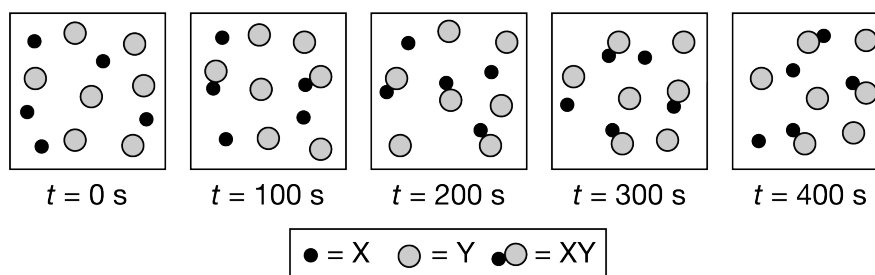
Time (minutes)	[X]	[XY]
0	5.0×10^{-2}	0.0
5	4.1×10^{-2}	9.0×10^{-3}
15	2.9×10^{-2}	2.1×10^{-2}
35	1.0×10^{-2}	4.0×10^{-2}
75	8.0×10^{-3}	4.2×10^{-2}
155	7.0×10^{-3}	4.3×10^{-2}
315	7.0×10^{-3}	4.3×10^{-2}
500	7.0×10^{-3}	4.3×10^{-2}



In an experiment, $\text{X}(g)$ and $\text{Y}(g)$ were combined in a rigid container at constant temperature and allowed to react as shown in the equation above. The table provides the data collected during the experiment. Based on the data, which of the following claims is most likely correct?

- (A) The reaction was about to reach equilibrium 15 minutes after the reactants were combined because the concentrations of X and XY were almost the same.
- (B) The reaction reached equilibrium between 75 minutes and 155 minutes after the reactants were combined because the concentrations of X and XY remained constant.
- (C) The reaction did not reach equilibrium because only 86% of the initial concentration of X was consumed.
- (D) The reaction did not reach equilibrium because initially there was no XY inside the container.

14.

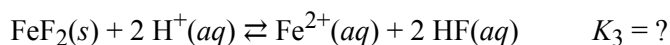
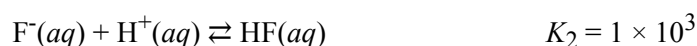
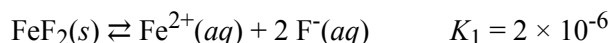


The particle diagram above illustrates the changes that take place when $\text{X}(g)$ and $\text{Y}(g)$ are placed inside a rigid container at constant temperature. Which of the following is a characteristic of a system at equilibrium that is best represented by the particle diagram?

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- (A) The particle diagram shows that initially the reaction proceeds to the right to form products, which is a characteristic of a system at equilibrium.
- (B) The particle diagram shows that after 200 s the rate of the reverse reaction is slower than the rate of the forward reaction, which is a characteristic of a system at equilibrium.
- (C) The particle diagram shows that after 200 s there are no observable changes in the amounts of reactants and products, which is a characteristic of a system at equilibrium.
- (D) The particle diagram shows that between 0 s and 200 s the rates of the forward and reverse reactions are the same, which is a characteristic of a system at equilibrium.

15.



On the basis of the information above, the dissolution of $\text{FeF}_2(s)$ in acidic solution is

- (A) thermodynamically favorable, because $K_2 > 1$
- (B) thermodynamically favorable, because $K_3 > 1$
- (C) not thermodynamically favorable, because $K_1 < 1$
- (D) not thermodynamically favorable, because $K_3 < 1$

16.

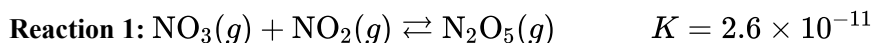
Reaction	Equilibrium Reactions at 298 K	K_{eq}
1	$\text{Br}_2(g) + \text{Cl}_2(g) \rightleftharpoons 2 \text{BrCl}(g)$	10.
2	$\text{N}_2(g) + \text{O}_2(g) \rightleftharpoons 2 \text{NO}(g)$	4.2×10^{-31}

The table above shows data for two reactions carried out in two separate evacuated 1.0-liter rigid containers at constant temperature of 298 K. To each container 0.50 mol of the appropriate reactants was added, and the reaction was allowed to reach equilibrium. Based on this information, which of the following correctly compares the relative concentrations of BrCl and NO present inside their respective containers at equilibrium?

- (A) $[\text{BrCl}]_{eq} = [\text{NO}]_{eq}$ because equimolecular mixtures of the reactants were allowed to reach equilibrium at the same constant temperature.
- (B) $[\text{BrCl}]_{eq} > [\text{NO}]_{eq}$ because Br_2 and Cl_2 are larger molecules that can collide more frequently to form products.
- (C) $[\text{BrCl}]_{eq} > [\text{NO}]_{eq}$ because the much larger K_{eq} for reaction 1 means that a much higher concentration of products will be present at equilibrium for reaction 1 compared with reaction 2.
- (D) $[\text{BrCl}]_{eq} < [\text{NO}]_{eq}$ because the much larger K_{eq} for reaction 1 means that hardly any products will be present at equilibrium compared with reaction 2.

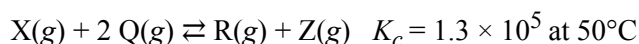
Chapter 10 Part 1: Keq

At night, an equilibrium reaction between two different nitrogen compounds generates N_2O_5 in the atmosphere, as represented below.



17. A mixture of NO_3 and a ten-fold excess of NO_2 are placed inside a rigid container at constant temperature and allowed to reach equilibrium. Which of the following provides a correct comparison of the equilibrium concentrations of these chemical species, and why?
- (A) $[\text{N}_2\text{O}_5] \ll [\text{NO}_3]$, because a small K value indicates that the consumption of the reactants is favored at equilibrium.
- (B) $[\text{NO}_2] \ll [\text{N}_2\text{O}_5]$, because a small K value indicates that the consumption of the reactants is favored at equilibrium.
- (C) $[\text{NO}_2] \ll [\text{NO}_3]$, because a small K value indicates that the formation of products is not favored at equilibrium.
- (D) $[\text{N}_2\text{O}_5] \ll [\text{NO}_2]$, because a small K value indicates that the formation of products is not favored at equilibrium.
-

18.



A 1.0 mol sample of $\text{X}(g)$ and a 1.0 mol sample of $\text{Q}(g)$ are introduced into an evacuated, rigid 10.0 L container and allowed to reach equilibrium at 50°C according to the equation above. At equilibrium, which of the following is true about the concentrations of the gases?

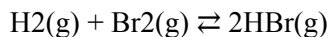
- (A) $[\text{R}] = 1/2[\text{Q}]$
- (B) $[\text{Q}] = 1/2[\text{X}]$
- (C) $[\text{R}] = [\text{Z}] > [\text{Q}]$
- (D) $[\text{x}] = [\text{Q}] = [\text{R}] = [\text{Z}]$
19. $2 \text{POCl}_3(g) \rightleftharpoons 2 \text{PCl}_3(g) + \text{O}_2(g) \quad \Delta G^\circ_{\text{rxn}} = +490 \text{ kJ/mol}$

A sample of $\text{POCl}_3(g)$ is placed in a closed, rigid container at 298 K and allowed to reach equilibrium according to the equation above. Based on the value for $\Delta G^\circ_{\text{rxn}} = +490 \text{ kJ/mol}$, which of the following is true?

- (A) $K = e^{-\frac{490,000}{8.314 \times 298}} \ll 1$ and at equilibrium $P_{\text{POCl}_3} \gg P_{\text{PCl}_3}$.
- (B) $K = e^{\frac{8.314 \times 298}{490,000}} > 1$ and at equilibrium $P_{\text{O}_2} > P_{\text{POCl}_3}$.
- (C) $K = e^{-\frac{490}{8.314 \times 298}} < 1$ and at equilibrium $P_{\text{POCl}_3} < P_{\text{PCl}_3}$.
- (D) $K = e^{\frac{8.314 \times 298}{490}} \gg 1$ and at equilibrium $P_{\text{O}_2} \gg P_{\text{POCl}_3}$.
-

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20.



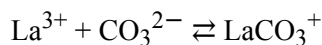
At a certain temperature, the value of the equilibrium constant, K , for the reaction represented above is 2.0×10^5 . What is the value of K for the reverse reaction at the same temperature?

- (A) -2.0×10^{-5}
- (B) 5.0×10^{-6}
- (C) 2.0×10^{-5}
- (D) 5.0×10^{-5}
- (E) 5.0×10^{-4}

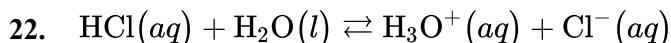
21.

Reaction	K_{eq}
$\text{La}^{3+} + \text{OH}^- + \text{HCO}_3^- \rightleftharpoons \text{LaCO}_3^+ + \text{H}_2\text{O}$	K_1
$\text{HCO}_3^- \rightleftharpoons \text{H}^+ + \text{CO}_3^{2-}$	K_a
$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$	K_w

Based on the information above, which of the following expressions represents the equilibrium constant, K , for the reaction represented by the equation below?



- (A) $K = (K_1)(K_a)(K_w)$
- (B) $K = (K_1)(K_a) / (K_w)$
- (C) $K = (K_1) / ((K_a)(K_w))$
- (D) $K = (K_1)(K_w) / (K_a)$

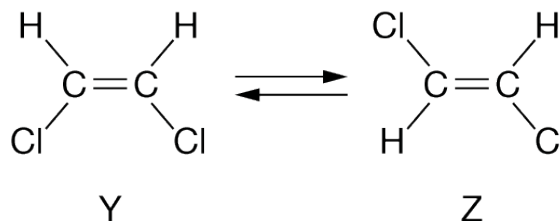


In $1.0 \text{ M HCl}(\text{aq})$, HCl is nearly 100 percent dissociated, as represented by the equation above. Which of the following best helps to explain why, in $0.01 \text{ M HCN}(\text{aq})$, less than 1 percent of HCN is dissociated?

- (A) The CN^- ion is not very soluble in water, and a solid precipitate would form if more of the HCN dissociated.
- (B) Compared to the $\text{HCl}(\text{aq})$ solution, the concentration of the $\text{HCN}(\text{aq})$ solution is much too dilute to achieve 100 percent dissociation.
- (C) The equilibrium constant for the dissociation of $\text{HCN}(\text{aq})$ is much smaller than that for the dissociation of $\text{HCl}(\text{aq})$.
- (D) $\text{HCN}(\text{aq})$ reacts with water to form a basic solution, and the high concentration of $\text{OH}^-(\text{aq})$ interferes with the dissociation process.

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23.



Substance	Y	Z
Initial concentration (M)	2.0	0.0
Equilibrium concentration (M)	1.2	0.80

The diagram above represents the equilibrium between the two isomers of $C_2H_2Cl_2$, and the table provides the data collected in an experiment to determine its equilibrium constant, K_c , at 490 K. In a second experiment done at the same temperature, $[Z]_{eq} \approx 1.0 M$. Which of the following is the approximate equilibrium concentration of Y in the second experiment, and why?

- (A) $[Y]_{eq} \approx 1.4 M$ because $[Y]_{eq} - [Z]_{eq}$ should be the same for the same reaction.
- (B) $[Y]_{eq} \approx 1.5 M$ because the ratio $\frac{[Z]_{eq}}{[Y]_{eq}}$ should remain constant when the reaction is done at the same temperature.
- (C) $[Y]_{eq} \approx 1.6 M$ because the ratio $\frac{[Y]_{initial}}{[Y]_{eq}}$ should remain constant when the reaction is done at the same temperature.
- (D) $[Y]_{eq} \approx 2.0 M$ because $\left([Y]_{initial} - [Y]_{eq}\right) = \left([Z]_{initial} - [Z]_{eq}\right)$ should be the same for the same reaction.

24. $2 A(g) + B(g) \rightleftharpoons 2 C(g)$

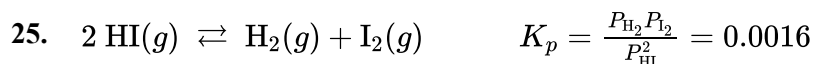
$A(g)$ and $B(g)$ react to form $C(g)$, according to the balanced equation above. In an experiment, a previously evacuated rigid vessel is charged with $A(g)$, $B(g)$, and $C(g)$, each with a concentration of $0.0100 M$. The following table shows the concentrations of the gases at equilibrium at a particular temperature.

$[A]_{eq}$	$[B]_{eq}$	$[C]_{eq}$
0.0180	0.0140	0.0020

If the experiment is repeated at a higher temperature at which K_c is larger, which of the following best describes the effect of the temperature change on the concentrations of the gases at equilibrium?

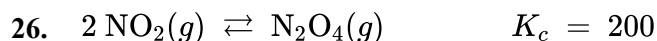
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- (A) $[A]_{eq}$, $[B]_{eq}$, and $[C]_{eq}$ will all increase because K_c increased.
- (B) $[A]_{eq}$ and $[B]_{eq}$ will remain constant, but $[C]_{eq}$ will increase because K_c increased.
- (C) There will be a decrease in $[A]_{eq}$ that will be two times the decrease in $[B]_{eq}$ because A and B react in a 2-to-1 ratio.
- (D) There will be an increase in $[A]_{eq}$ that will be two times the increase in $[B]_{eq}$ because A and B react in a 2-to-1 ratio.

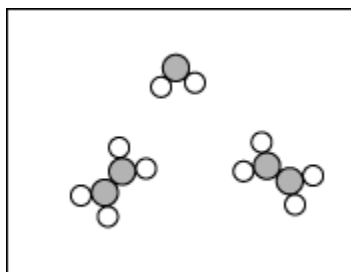


The decomposition of $\text{HI}(g)$ at 298 K is represented by the equilibrium equation above. When 100. torr of $\text{HI}(g)$ is added to a previously evacuated, rigid container and allowed to reach equilibrium, the partial pressure of $\text{I}_2(g)$ is approximately 3.7 torr. If the initial pressure of $\text{HI}(g)$ is increased to 200. torr and the process is repeated at the same temperature, which of the following correctly predicts the equilibrium partial pressure of $\text{I}_2(g)$, and why?

- (A) $P_{\text{I}_2} \approx 14$ torr, because it is directly proportional to the square of the initial pressure of HI .
- (B) $P_{\text{I}_2} \approx 0.073$ torr, because it is inversely proportional to the square of the initial pressure of HI .
- (C) $P_{\text{I}_2} \approx 7.4$ torr, because it is directly proportional to the initial pressure of HI .
- (D) $P_{\text{I}_2} \approx 1.9$ torr, because it is inversely proportional to the initial pressure of HI .



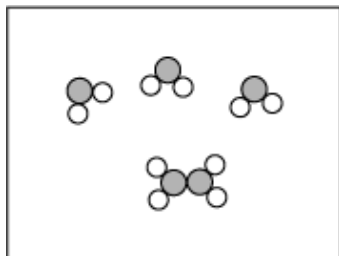
The equilibrium system shown above is represented in the diagram below in which each molecule represents one mole of the substance in a 100. L container at a constant temperature.



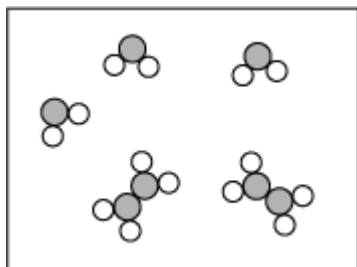
Additional $\text{N}_2\text{O}_4(g)$ is added to the container and allowed to reestablish equilibrium. Which of the following best represents the system after the equilibrium has been reestablished?

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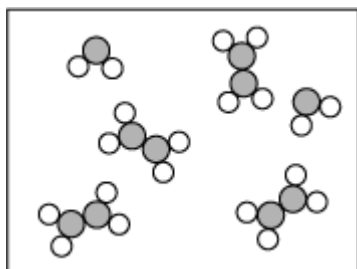
(A)



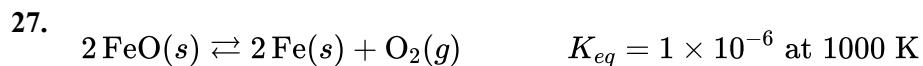
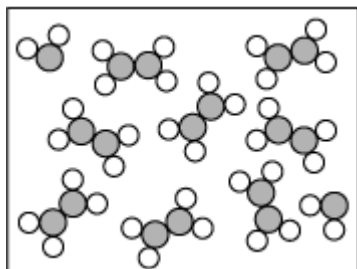
(B)



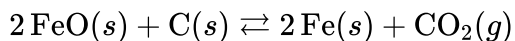
(C)



(D)



The formation of $\text{Fe}(s)$ and $\text{O}_2(g)$ from $\text{FeO}(s)$ is not thermodynamically favorable at room temperature. In an effort to make the process favorable, $\text{C}(s)$ is added to the $\text{FeO}(s)$ at elevated temperatures. Based on the information above, which of the following gives the value of K_{eq} and the sign of ΔG° for the reaction represented by the equation below at 1000 K?



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(A)

K_{eq}	ΔG°
1×10^{-38}	Positive

(B)

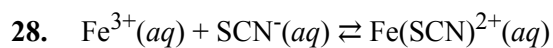
K_{eq}	ΔG°
1×10^{-38}	Negative

(C)

K_{eq}	ΔG°
1×10^{26}	Positive

(D)

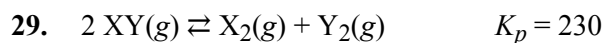
K_{eq}	ΔG°
1×10^{26}	Negative



For the reaction represented above, the value of the equilibrium constant, K_{eq} , is 240 at 25°C. From this information, correct deductions about the reaction at 25°C include which of the following?

- I. The reaction is quite rapid.
 - II. The product is favored over the reactants at equilibrium.
 - III. The reaction is endothermic.
- (A) I only
(B) II only
(C) I and II only
(D) II and III only
(E) I, II, and III

Chapter 10 Part 1: Keq



A certain gas, $\text{XY}(g)$, decomposes as represented by the equation above. A sample of each of the three gases is put in a previously evacuated container. The initial partial pressures of the gases are shown in the table below.

Gas	Initial Partial Pressure (atm)
XY	0.010
X_2	0.20
Y_2	2.0

The temperature of the reaction mixture is held constant. In which direction will the reaction proceed?

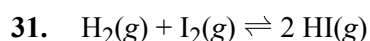
- (A) The reaction will form more products.
- (B) The reaction will form more reactant.
- (C) The mixture is at equilibrium, so there will be no change.
- (D) It cannot be determined unless the volume of the container is known.

30.

Reaction	K_{eq}
$\text{SO}_2(g) + \frac{1}{2} \text{O}_2(g) \rightleftharpoons \text{SO}_3(g)$	K_1
$2 \text{SO}_3(g) \rightleftharpoons 2 \text{SO}_2(g) + \text{O}_2(g)$	K_2

Which of the following shows the relationship between K_1 and K_2 in the reactions represented above?

- (A) $K_2 = (K_1)^2$
- (B) $K_2 = \sqrt{K_1}$
- (C) $K_2 = \frac{1}{(K_1)^2}$
- (D) $K_2 = \frac{1}{K_1}$



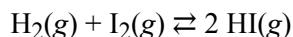
$K_c = 50$. at 600°C

Equimolar samples of each of three gases, $\text{H}_2(g)$, $\text{I}_2(g)$, and $\text{HI}(g)$, are introduced into a 3.0 L container that is heated to 600°C . According to the information above, which of the following will occur as the system approaches equilibrium?

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- (A) More $\text{H}_2(\text{g})$ will be produced because $Q > K_c$.
- (B) More $\text{I}_2(\text{g})$ will be produced because $Q < K_c$.
- (C) More $\text{HI}(\text{g})$ will be produced because $Q < K_c$.
- (D) No change will occur because $Q = K_c$.

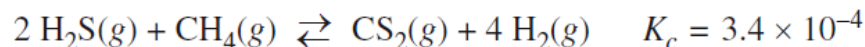
32.



At 450°C , 2.0 moles each of $\text{H}_2(\text{g})$, $\text{I}_2(\text{g})$, and $\text{HI}(\text{g})$ are combined in a 1.0 L rigid container. The value of K_c at 450°C is 50. Which of the following will occur as the system moves toward equilibrium?

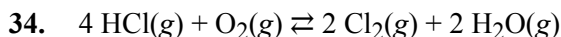
- (A) More $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ will form.
- (B) More $\text{HI}(\text{g})$ will form.
- (C) The total pressure will decrease.
- (D) No net reaction will occur, because the number of molecules is the same on both sides of the equation.

33.



A 0.10 mol sample of each of the four species in the reaction represented above is injected into a rigid, previously evacuated 1.0 L container. Which of the following species will have the highest concentration when the system reaches equilibrium?

- (A) $\text{H}_2\text{S}(\text{g})$
- (B) $\text{CH}_4(\text{g})$
- (C) $\text{CS}_2(\text{g})$
- (D) $\text{H}_2(\text{g})$



Equal numbers of moles of HCl and O_2 in a closed system are allowed to reach equilibrium as represented by the equation above. Which of the following must be true at equilibrium?

I. $[\text{HCl}]$ must be less than $[\text{Cl}_2]$.

II. $[\text{O}_2]$ must be greater than $[\text{HCl}]$.

III. $[\text{Cl}_2]$ must equal $[\text{H}_2\text{O}]$.

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only
- (E) I, II, and III

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35.

$\text{H}_2\text{CO}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HCO}_3^-(aq) + \text{H}_3\text{O}^+(aq)$	K_1
$\text{HCO}_3^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{CO}_3^{2-}(aq) + \text{H}_3\text{O}^+(aq)$	K_2
$\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$	K_3
$2 \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OH}^-(aq)$	K_4

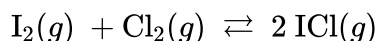


The table above lists some equilibrium systems and their equilibrium constants. Which of the following identifies the correct mathematical relationship that uses the information to calculate K_5 ?

- (A) $K_5 = \frac{K_1 \times K_3}{K_4}$
 (B) $K_5 = \frac{K_4}{K_1 \times K_3}$
 (C) $K_5 = K_1 \times K_3 \times K_4$
 (D) $K_5 = \frac{K_1 \times K_3}{K_2}$

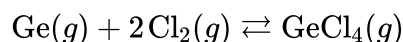


Based on the information above, what is the value of K_{eq} for the reaction represented below?



- (A) 0.01
 (B) 10
 (C) 20
 (D) 100

37.

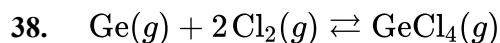


The value of the equilibrium constant for the reaction represented above is 1×10^{10} . What is the value of the equilibrium constant for the following reaction?

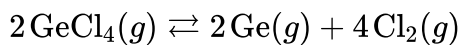


- (A) 1×10^{-20}
 (B) 1×10^{-10}
 (C) 1×10^{10}
 (D) 1×10^{20}

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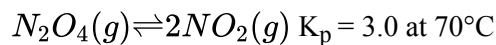


The value of the equilibrium constant for the reaction represented above is 2.3×10^{11} . What is the value of the equilibrium constant for the following reaction?



- (A) 1.9×10^{-23}
(B) 4.3×10^{-12}
(C) 2.3×10^{11}
(D) 5.3×10^{22}

39.

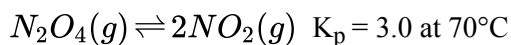


colorless brown

A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C , as represented above.

If $P_{\text{N}_2\text{O}_4}$ is 0.64 atm when the system is at equilibrium at 70°C , what is P_{NO_2} ?

- (A) 0.21 atm
(B) 1.4 atm
(C) 1.9 atm
(D) 4.7 atm
-



colorless brown

A mixture of $\text{NO}_2(g)$ and $\text{N}_2\text{O}_4(g)$ is placed in a glass tube and allowed to reach equilibrium at 70°C , as represented above.

40. If $P_{\text{N}_2\text{O}_4}$ is 1.33 atm when the system is at equilibrium at 70°C , what is P_{NO_2} ?

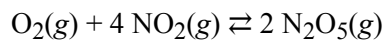
- (A) 0.44 atm
(B) 2.0 atm
(C) 2.3 atm
(D) 4.0 atm
-

Chapter 10 Part 1: Keq

41.

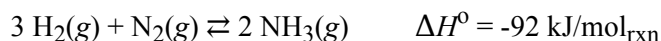


The equilibrium constant for the gas phase reaction above is 95 at 25°C. What is the value of the equilibrium constant for the following reaction at 25°C?

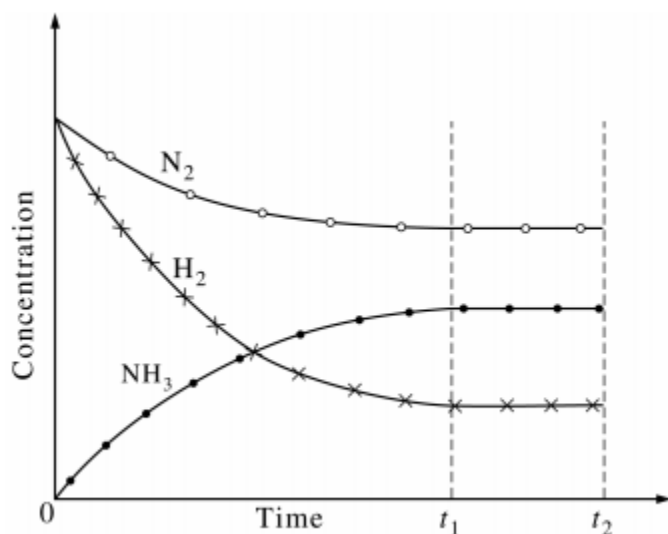


- (A) $(95)^2$
(B) 95
(C) $(95)^{\frac{1}{2}}$
(D) $\frac{1}{95}$
(E) $\frac{1}{(95)^2}$

H_2 gas and N_2 gas were placed in a rigid vessel and allowed to reach equilibrium in the presence of a catalyst according to the following equation.



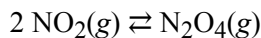
The diagram below shows how the concentrations of H_2 , N_2 , and NH_3 in this system changed over time.



42. Which of the following was true for the system between time t_1 and time t_2 ?

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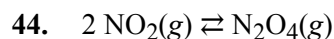
- (A) The concentration of N_2 decreased.
 - (B) The temperature of the system decreased.
 - (C) The number of effective collisions between H_2 and N_2 was zero.
 - (D) The rates of the forward and reverse reactions were equal.
 - (E) The rate of formation of NH_3 molecules was equal to the rate of disappearance of H_2 molecules.
-



dark brown colorless

The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

43. A 0.0300 mol sample of $\text{NO}_2(g)$ is placed in a rigid 1.00 L reaction vessel and allowed to reach equilibrium at a certain temperature. What is the value of K_c at this temperature if 0.00500 mol of $\text{N}_2\text{O}_4(g)$ is present at equilibrium?
- (A) 0.0800
 - (B) 0.250
 - (C) 4.00
 - (D) 12.5
-



dark brown colorless

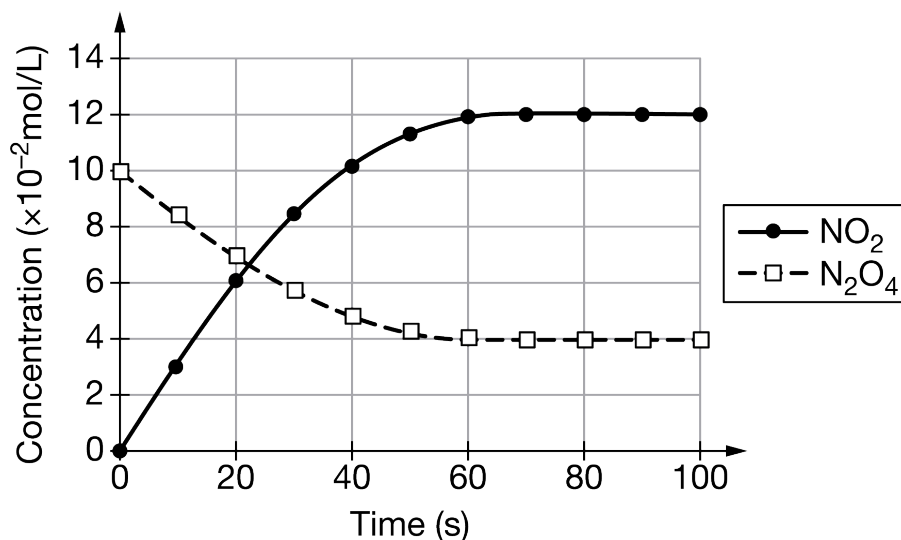
The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

A 0.0688 mol sample of $\text{NO}_2(g)$ is placed in a rigid 1.00 L reaction vessel and allowed to reach equilibrium at a certain temperature. What is the value of K_c at this temperature if 0.0133 mol of $\text{N}_2\text{O}_4(g)$ is present at equilibrium?

- (A) 0.134
 - (B) 0.315
 - (C) 4.32
 - (D) 7.47
-

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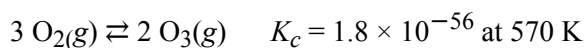
45.



A sample of $\text{N}_2\text{O}_4(g)$ is placed into an evacuated container at 373 K and allowed to undergo the reversible reaction $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$. The concentration of each species is measured over time, and the data are used to make the graph shown above. Which of the following identifies when equilibrium is first reached and provides a correct explanation?

- (A) At 14 seconds, because $[\text{N}_2\text{O}_4]$ is twice $[\text{NO}_2]$, which implies that the forward and reverse reaction rates are equal.
- (B) At 23 seconds, because $[\text{NO}_2]$ equals $[\text{N}_2\text{O}_4]$, which shows that equal concentrations are present at equilibrium.
- (C) At 40 seconds, because $[\text{NO}_2]$ is twice $[\text{N}_2\text{O}_4]$, which matches the stoichiometry of the balanced chemical equation.
- (D) At 60 seconds, because $[\text{NO}_2]$ and $[\text{N}_2\text{O}_4]$ remain constant, indicating that the forward and reverse reaction rates are equal.

46.

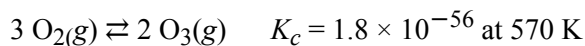


For the system represented above, $[\text{O}_2]$ and $[\text{O}_3]$ initially are 0.150 mol/L and 2.5 mol/L respectively. Which of the following best predicts what will occur as the system approaches equilibrium at 570 K?

- (A) The amount of $\text{O}_3(g)$ will increase, because $Q < K_c$.
- (B) The amount of $\text{O}_3(g)$ will decrease, because $Q < K_c$.
- (C) The amount of $\text{O}_3(g)$ will increase, because $Q > K_c$.
- (D) The amount of $\text{O}_3(g)$ will decrease, because $Q > K_c$.

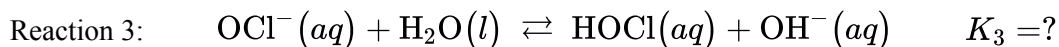
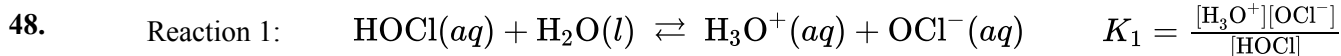
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47.



For the system represented above, $[\text{O}_2]$ and $[\text{O}_3]$ initially are 0.131 mol/L and 0.789 mol/L respectively. Which of the following best predicts what will occur as the system approaches equilibrium at 570 K?

- (A) The amount of $\text{O}_3(g)$ will increase, because $Q < K_c$.
- (B) The amount of $\text{O}_3(g)$ will decrease, because $Q < K_c$.
- (C) The amount of $\text{O}_3(g)$ will increase, because $Q > K_c$.
- (D) The amount of $\text{O}_3(g)$ will decrease, because $Q > K_c$.



Based on the equilibrium constants given above, which of the following gives the correct expression for the equilibrium constant for reaction 3?

- (A) $K_3 = \frac{K_2}{K_1}$
- (B) $K_3 = K_1 K_2$
- (C) $K_3 = \frac{K_1}{K_2}$
- (D) $K_3 = \frac{1}{K_1 K_2}$

Refer to the following.



$\text{PCl}_5(g)$ decomposes into $\text{PCl}_3(g)$ and $\text{Cl}_2(g)$ according to the equation above. A pure sample of $\text{PCl}_5(g)$ is placed in a rigid, evacuated 1.00 L container. The initial pressure of the $\text{PCl}_5(g)$ is 1.00 atm. The temperature is held constant until the $\text{PCl}_5(g)$ reaches equilibrium with its decomposition products. The figures below show the initial and equilibrium conditions of the system.

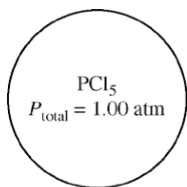


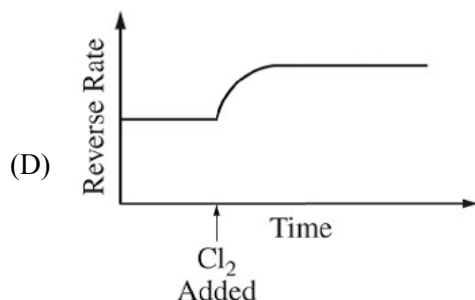
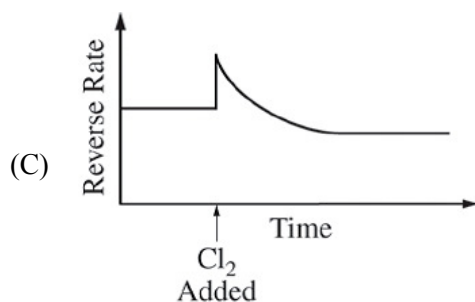
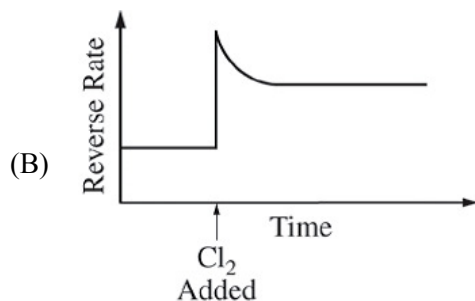
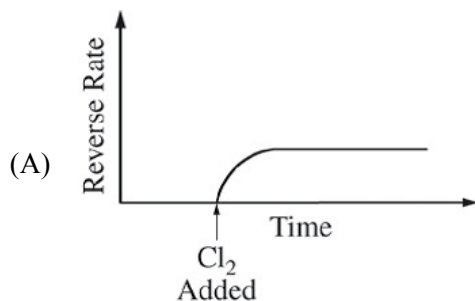
Figure 1: Initial



Figure 2: Equilibrium

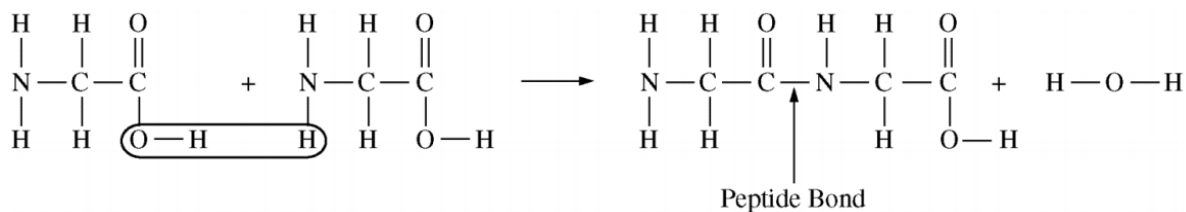
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49. As the reaction progresses toward equilibrium, the rate of the forward reaction
- (A) increases until it becomes the same as the reverse reaction rate at equilibrium
 - (B) stays constant before and after equilibrium is reached
 - (C) decreases to become a constant nonzero rate at equilibrium
 - (D) decreases to become zero at equilibrium
50. Additional $\text{Cl}_2(\text{g})$ is injected into the system at equilibrium. Which of the following graphs best shows the rate of the reverse reaction as a function of time? (Assume that the time for injection and mixing of the additional $\text{Cl}_2(\text{g})$ is negligible.)



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51. Which of the following statements about K_p , the equilibrium constant for the reaction, is correct?
- (A) $K_p > 1$
 (B) $K_p < 1$
 (C) $K_p = 1$
 (D) It cannot be determined whether $K_p > 1$, $K_p < 1$, or $K_p = 1$ without additional information.



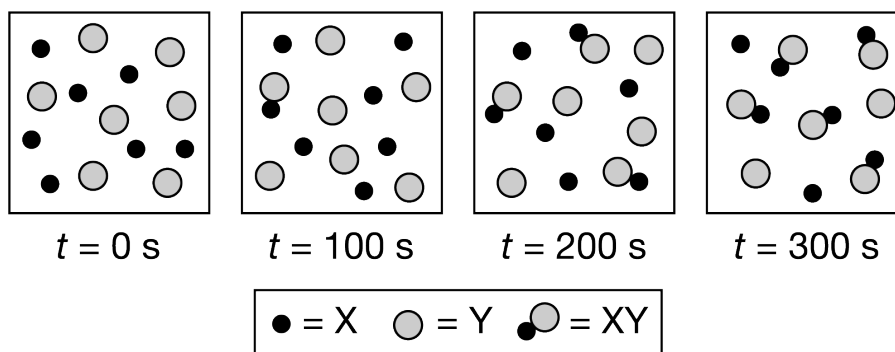
Two molecules of the amino acid glycine join through the formation of a peptide bond, as shown above. The thermodynamic data for the reaction are listed in the following table.

ΔG_{298}°	ΔH_{298}°	ΔS_{298}°
+15 kJ/mol _{rxn}	+12 kJ/mol _{rxn}	-10 J/(K·mol _{rxn})

52. Based on the thermodynamic data, which of the following is true at 298 K?
- (A) $K_{eq} = 0$
 (B) $0 < K_{eq} < 1$
 (C) $K_{eq} = 1$
 (D) $K_{eq} > 1$

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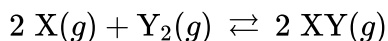
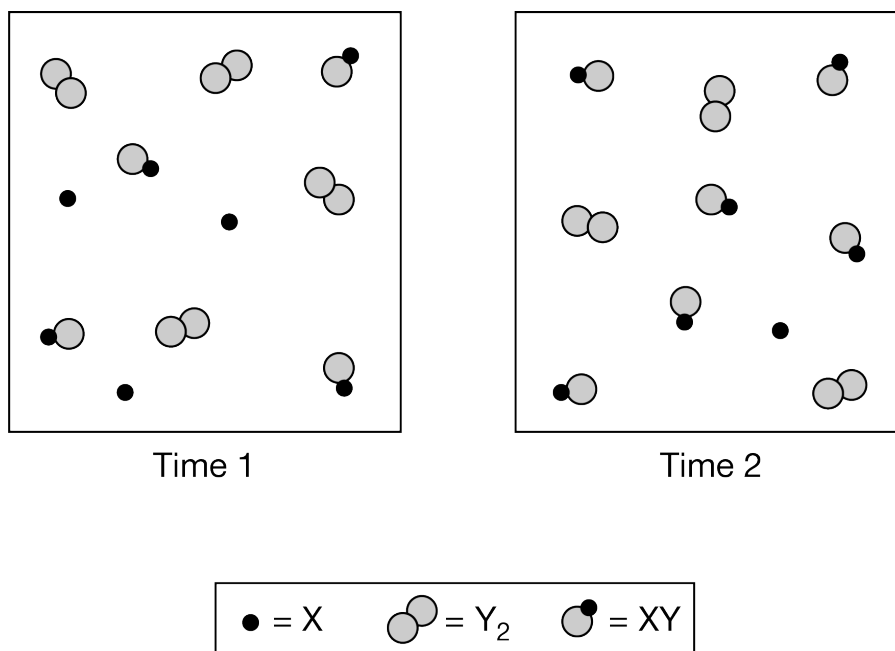
53.



An equimolar mixture of $X(g)$ and $Y(g)$ is placed inside a rigid container at constant temperature. The particle diagram above represents the changes that occur over time. Based on the particle diagram, which of the following best predicts whether or not the system has reached equilibrium by 300 s?

- (A) It is not possible to determine that the system has reached equilibrium by 300 s because the stoichiometry of the reaction is not known.
- (B) It is not possible to determine that the system has reached equilibrium by 300 s because the amounts of X, Y, and XY have continued to change.
- (C) The system has reached equilibrium by 300 s because the rate of formation of XY is constant.
- (D) The system has reached equilibrium by 300 s because the rates of consumption of X and Y are equal.

54.



A reversible reaction is represented by the equation above. The amounts of reactants and products at time 1 are shown in the particle diagram on the left. The particle diagram on the right shows the amounts of reactants and products at time 2. Based on the diagrams, what can be inferred about the relative rates of the forward and reverse reactions between time 1 and time 2?

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- (A) Nothing can be inferred because the total number of X and Y atoms is the same in each diagram.
- (B) Nothing can be inferred because the temperature of the system may have been changed.
- (C) The rate of the reverse reaction is greater than the rate of the forward reaction.
- (D) The rate of the forward reaction is greater than the rate of the reverse reaction.

55.



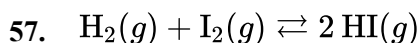
At a certain point in time, a 1.00 L rigid reaction vessel contains 1.5 mol of $\text{PCl}_3(g)$, 1.0 mol of $\text{Cl}_2(g)$, and 2.5 mol of $\text{PCl}_5(g)$. Which of the following describes how the measured pressure in the reaction vessel will change and why it will change that way as the reaction system approaches equilibrium at constant temperature?

- (A) The pressure will increase because $Q < K_c$.
- (B) The pressure will increase because $Q > K_c$.
- (C) The pressure will decrease because $Q < K_c$.
- (D) The pressure will decrease because $Q > K_c$.



Equal volumes of 0.1 M $\text{AgNO}_3(aq)$ and 0.4 M $\text{NH}_3(aq)$ are mixed and allowed to reach the equilibrium represented above. Which of the following correctly lists the equilibrium concentrations of the $\text{Ag}^+(aq)$, $\text{NH}_3(aq)$, and $[\text{Ag}(\text{NH}_3)_2]^+(aq)$ in order from least to greatest?

- (A) $[\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
- (B) $[\text{NH}_3]_{eq} < [\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
- (C) $[[\text{Ag}(\text{NH}_3)_2]^+]_{eq} < [\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq}$
- (D) $[\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq} < [\text{NH}_3]_{eq}$

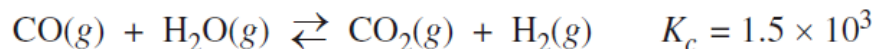


Hydrogen gas reacts with iodine gas at constant temperature in a sealed rigid container. The gases are allowed to reach equilibrium according to the equation above. Which of the following best describes what will happen to the reaction immediately after additional iodine gas is added to the system?

- (A) The rates of both the forward and reverse reactions decrease.
- (B) The rates of both the forward and reverse reactions do not change.
- (C) The rate of the forward reaction becomes greater than the rate of the reverse reaction.
- (D) The rate of the forward reaction becomes less than the rate of the reverse reaction.

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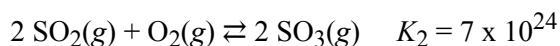
58.



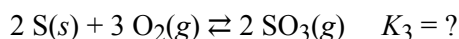
A 2.0 mol sample of $\text{CO}(g)$ and a 2.0 mol sample of $\text{H}_2\text{O}(g)$ are introduced into a previously evacuated 100. L rigid container, and the temperature is held constant as the reaction represented above reaches equilibrium. Which of the following is true at equilibrium?

- (A) $[\text{H}_2\text{O}] > [\text{CO}]$ and $[\text{CO}_2] > [\text{H}_2]$
- (B) $[\text{H}_2\text{O}] > [\text{H}_2]$
- (C) $[\text{CO}_2] > [\text{CO}]$
- (D) $[\text{CO}] = [\text{H}_2\text{O}] = [\text{CO}_2] = [\text{H}_2]$

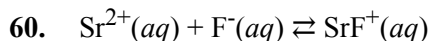
59.



Given the value of the equilibrium constants K_1 and K_2 for the reactions represented above, what is the value of the equilibrium constant, K_3 , for the following reaction?



- (A) 1×10^{130}
- (B) 3×10^{80}
- (C) 1×10^{65}
- (D) 2×10^{40}
- (E) 7×10^{24}

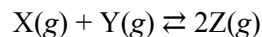


At 25°C , the equilibrium constant for the reaction represented above has a value of 1.3. At 50°C , the value of the equilibrium constant is less than 1.3. Based on this information, which of the following must be correct?

- (A) The reaction rate decreases as the temperature is increased.
- (B) The reaction is thermodynamically favorable only at temperatures above 25°C .
- (C) At 25°C , ΔG° for the reaction is positive.
- (D) At 25°C , ΔS° for the reaction is positive.
- (E) At 25°C , ΔH° for the reaction is negative.

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61.



When 4.00 mol each of X(g) and Y(g) are placed in a 1.00 L vessel and allowed to react at constant temperature according to the equation above, 6.00 mol of Z(g) is produced. What is the value of the equilibrium constant, K_c ?

- (A) 3
- (B) 6
- (C) 8
- (D) 16
- (E) 36

62. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$

A 1.0 mol sample of $\text{H}_2(\text{g})$ and a 1.0 mol sample of $\text{I}_2(\text{g})$ are placed in a previously evacuated 1.0 L container and allowed to reach the equilibrium shown above. If the equilibrium concentration of $\text{HI}(\text{g})$ is 0.40 M, which of the following is the value of the equilibrium constant?

- (A) 0.16
- (B) 0.25
- (C) 0.44
- (D) 1.0

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63. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



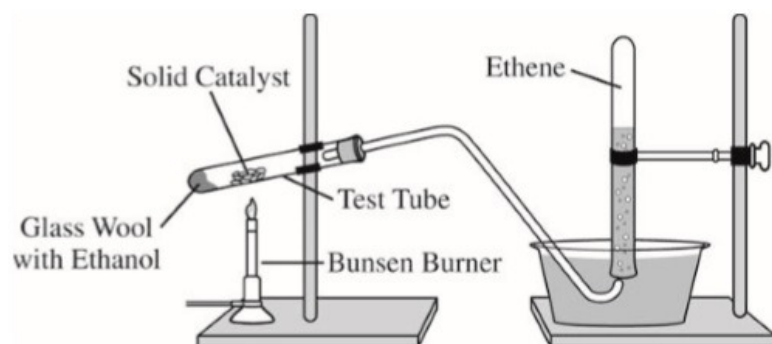
When heated strongly, solid calcium carbonate decomposes to produce solid calcium oxide and carbon dioxide gas, as represented by the equation above. A 2.0 mol sample of $\text{CaCO}_3(s)$ is placed in a rigid 100. L reaction vessel from which all the air has been evacuated. The vessel is heated to 898°C at which time the pressure of $\text{CO}_2(g)$ in the vessel is constant at 1.00 atm, while some $\text{CaCO}_3(s)$ remains in the vessel.

- (a) Calculate the number of moles of $\text{CO}_2(g)$ present in the vessel at equilibrium.
- (b) Write the expression for K_p , the equilibrium constant for the reaction, and determine its value at 898°C .
- (c) The experiment was repeated, but this time starting with a 4.0 mol sample of $\text{CaCO}_3(s)$. On the following graph, draw a curve showing how the pressure of $\text{CO}_2(g)$ would change over time as the vessel is heated to 898°C and equilibrium is established.

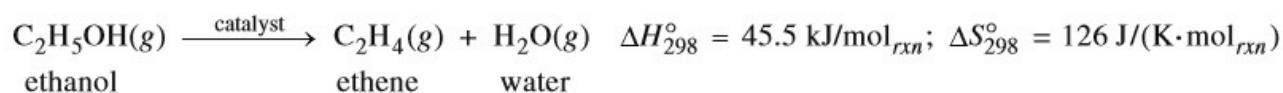


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64.



Ethene, $\text{C}_2\text{H}_4(\text{g})$ (molar mass 28.1 g/mol), may be prepared by the dehydration of ethanol, $\text{C}_2\text{H}_5\text{OH}(\text{g})$ (molar mass 46.1 g/mol), using a solid catalyst. A setup for the lab synthesis is shown in the diagram above. The equation for the dehydration reaction is given below.

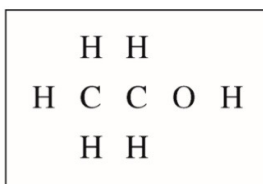
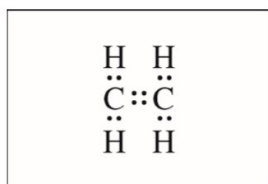


A student added a 0.200 g sample of $\text{C}_2\text{H}_5\text{OH}(\text{l})$ to a test tube using the setup shown above. The student heated the test tube gently with a Bunsen burner until all of the $\text{C}_2\text{H}_5\text{OH}(\text{l})$ evaporated and gas generation stopped. When the reaction stopped, the volume of collected gas was 0.0854 L at 0.822 atm and 305 K. (The vapor pressure of water at 305 K is 35.7 torr.)

- a. Calculate the number of moles of $\text{C}_2\text{H}_4(\text{g})$
 - i. that are actually produced in the experiment and measured in the gas collection tube and
 - ii. that would be produced if the dehydration reaction went to completion.
- b. Calculate the percent yield of $\text{C}_2\text{H}_4(\text{g})$ in the experiment.

Because the dehydration reaction is not observed to occur at 298 K, the student claims that the reaction has an equilibrium constant less than 1.00 at 298 K.

- c. Do the thermodynamic data for the reaction support the student's claim? Justify your answer, including a calculation of ΔG_{298}° for the reaction.
- d. The Lewis electron-dot diagram for C_2H_4 is shown below in the box on the left. In the box on the right, complete the Lewis electron-dot diagram for $\text{C}_2\text{H}_5\text{OH}$ by drawing in all of the electron pairs.

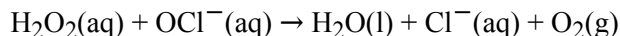


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- e. What is the approximate value of the C–O–H bond angle in the ethanol molecule?
- f. During the dehydration experiment, $\text{C}_2\text{H}_4(g)$ and unreacted $\text{C}_2\text{H}_5\text{OH}(g)$ passed through the tube into the water. The C_2H_4 was quantitatively collected as a gas, but the unreacted $\text{C}_2\text{H}_5\text{OH}$ was not. Explain this observation in terms of the intermolecular forces between water and each of the two gases.

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65.



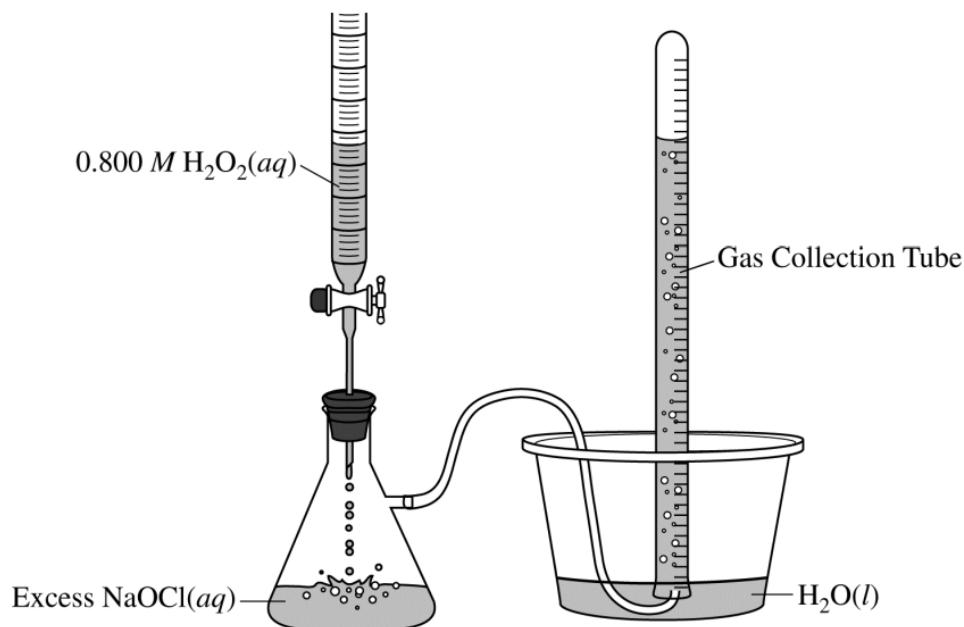
A student investigates the reaction between $\text{H}_2\text{O}_2(\text{aq})$ and $\text{NaOCl}(\text{aq})$, which is represented by the net-ionic equation shown above.

- a. Is the reaction represented above a redox reaction? Justify your answer.

To better understand the reaction, the student looks up thermodynamic data for the reaction. For the reaction represented above, ΔG_{298}° is $-197 \text{ kJ/mol}_{\text{rxn}}$ represented above, the value of and the value of ΔS_{298}° is $144 \text{ J/(K mol}_{\text{rxn}})$

- b. Calculate the value of ΔH_{298}° for the reaction in $\text{kJ/mol}_{\text{rxn}}$.
- c. Does the temperature inside the flask increase, decrease, or remain the same as the reaction proceeds? Justify your answer.
- d. Calculate the value of the equilibrium constant, K , for the reaction at 298 K

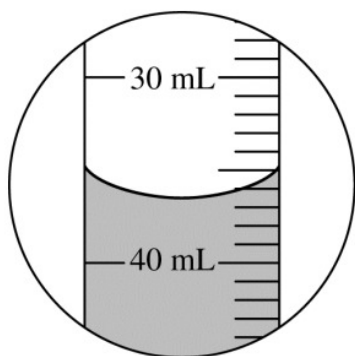
The student decides to produce 40.0 mL of $\text{O}_2(\text{g})$ at a pressure of 0.988 atm and a temperature of 298 K using the reaction represented above. The student uses the equipment shown below. The student sets up a 250 mL Erlenmeyer flask fitted with a one-hole stopper. The flask is connected to a 50 mL gas-collection tube that initially is completely filled with water.



- e. Calculate the volume of 0.800 M $\text{H}_2\text{O}_2(\text{aq})$ that the student should add to excess $\text{NaOCl}(\text{aq})$ to produce 40.0 mL of $\text{O}_2(\text{g})$ at 0.988 atm and 298 K.
- f. The student added the amount of $\text{H}_2\text{O}_2(\text{aq})$ calculated in part (e) to excess $\text{NaOCl}(\text{aq})$. However, instead

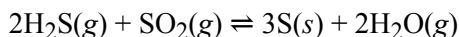
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of producing 40.0 mL of $\text{O}_2(\text{g})$, the volume indicated in the diagram below was produced.



- Based on the diagram above, what volume of gas was produced?
 - Assuming that all the gas in the tube is $\text{O}_2(\text{g})$, calculate the percent yield of $\text{O}_2(\text{g})$.
 - Is the assumption that all the gas in the tube is $\text{O}_2(\text{g})$ correct? Explain.
- g. To account for the percent yield being less than 100 percent, the student claims that the reaction reached equilibrium before the expected amount of $\text{O}_2(\text{g})$ was produced. Considering your answer to part (d) above, do you agree or disagree with the student's claim? Justify your answer.

66.



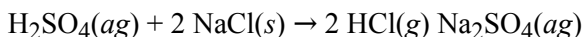
At 298 K, the standard enthalpy change, ΔH° , for the reaction represented above is -145 kilojoules.

- Predict the sign of the standard entropy change, ΔS° , for the reaction. Explain the basis for your prediction.
- At 298 K, the forward reaction (i.e., toward the right) is spontaneous. What change, if any, would occur in the value of ΔG° for this reaction as the temperature is increased? Explain your reasoning using thermodynamic principles.
- What change, if any, would occur in the value of the equilibrium constant, K_{eq} , for the situation described in (b)? Explain your reasoning.
- The absolute temperature at which the forward reaction becomes nonspontaneous can be predicted. Write the equation that is used to make the prediction. Why does this equation predict only an approximate value for the temperature?

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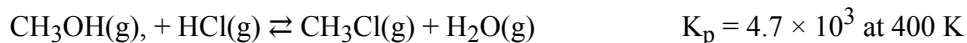
67. Answer the following questions relating to HCl, CH₃Cl, and CH₃Br.

- a. HCl(g), can be prepared by the reaction of concentrated H₂SO₄(ag), with NaCl(s), as represented by the following equation.



- A student claims that the reaction is a redox reaction. Is the student correct? Justify your answer.
- Calculate the mass, in grams, of NaCl(s), needed to react with excess H₂SO₄(ag) to produce 3.0 grams of HCl(g). Assume that the reaction goes to completion.

HCl(g), can react with methanol vapor, CH₃OH(g), to produce CH₃Cl(g), as represented by the following equation.



- CH₃OH(g), and HCl(g), are combined in a 10.0 liter sealed reaction vessel and allowed to reach equilibrium at 400 K. The initial partial pressure of CH₃OH(g) in the vessel is 0.250 atm and that of HCl(g) is 0.600 atm.
 - Does the total pressure in the vessel increase, decrease, or remain the same as equilibrium is approached? Justify your answer in terms of the reaction stoichiometry.
 - Considering the value of K_p , calculate the final partial pressure of HCl(g) after the system inside the vessel reaches equilibrium at 400 K.
 - The student claims that the final partial pressure of CH₃OH(g) at equilibrium is very small but not exactly zero. Do you agree or disagree with the student's claim? Justify your answer.
- The table below shows some data for the compounds CH₃Cl and CH₃Br.

Compound	Molar Mass (g/mol)	Dipole Moment (D)	Boiling Point (K)	$\Delta H_{\text{vaporization}}^\circ$ (kJ/mol)
CH ₃ Cl	50.5	1.87	249	18.9
CH ₃ Br	94.9	1.81	277	22.8

- Identify all the types of intermolecular forces that exist among molecules in CH₃Cl(l).
- In terms of intermolecular forces, explain why the boiling point of CH₃Br(l) is greater than that

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of $\text{CH}_3\text{Cl}(l)$.

- d. A 2.00 milliliter sealed glass vial containing a 1.00 gram sample of $\text{CH}_3\text{Cl}(l)$ is stored in a freezer at 233 K.
- Calculate the pressure in the vial at 298 K assuming that all the $\text{CH}_3\text{Cl}(l)$ vaporizes.
 - Explain why it would be unsafe to remove the vial from the freezer and leave it on a lab bench at 298 K.

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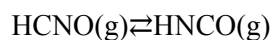
68. Answer the following questions about the isomers fulminic acid and isocyanic acid.

Two possible Lewis electron-dot diagrams for fulminic acid, HCNO, are shown below.



- a. Explain why the diagram on the left is the better representation for the bonding in fulminic acid. Justify your choice based on formal charges.

Fulminic acid can convert to isocyanic acid according to the equation below.



fulminic acid - isocyanic acid

Fulminic Acid	Isocyanic Acid
$\text{H}-\text{C}\equiv\text{N}-\ddot{\text{O}}:$	$\text{H}-\ddot{\text{N}}=\text{C}=\ddot{\text{O}}:$

- b. Using the Lewis electron-dot diagrams of fulminic acid and isocyanic acid shown in the boxes above and the table of average bond enthalpies below, determine the value of ΔH° for the reaction of HCNO(g) to

form HNCO(g).	Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)
	N-O	201		C=N	615		H-C	413
	C=O	745		C≡N	891		H-N	391

- c. A student claims that ΔS° for the reaction is close to zero. Explain why the student's claim is accurate.
- d. Which species, fulminic acid (HCNO) or isocyanic acid (HNCO), is present in higher concentration at equilibrium at 298 K? Justify your answer in terms of thermodynamic favorability and the equilibrium constant.

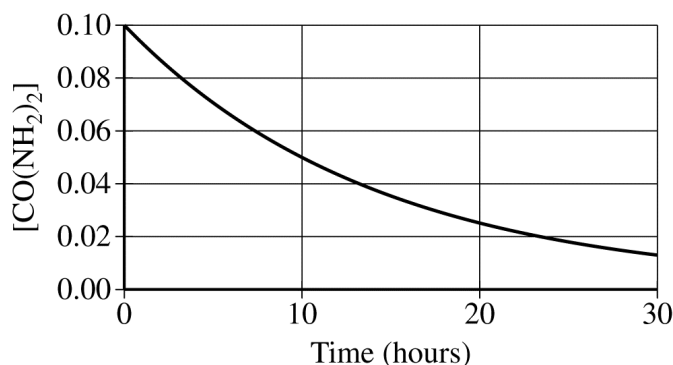
The ammonium salt of isocyanic acid is a product of the decomposition of urea, $\text{CO}(\text{NH}_2)_2$, represented below.



A student studying the decomposition reaction runs the reaction at 90°C. The student collects data on the concentration of urea as a function of time, as shown by the data table and the graph below.

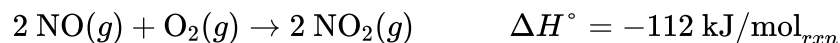
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Time (hours)	[CO(NH ₂) ₂]
0	0.1000
5	0.0707
10	0.0500
15	0.0354
20	0.0250
25	0.0177
30	0.0125



- e. The student proposes that the rate law is $rate = k[CO(NH_2)_2]$.
- Explain how the data support the student's proposed rate law.
 - Using the proposed rate law and the student's results, determine the value of the rate constant, k . Include units with your answer.
- f. The student learns that the decomposition reaction was run in a solution with a pH of 13. Briefly describe an experiment, including the initial conditions that you would change and the data you would gather, to determine whether the rate of the reaction depends on the concentration of $OH^-(aq)$.

69. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Answer the following questions about the reaction represented above at 298 K.

	Absolute Entropy at 298 K (J/(K mol))
NO(g)	211
O ₂ (g)	205
NO ₂ (g)	240

- Using the information in the table above, calculate the value of ΔS° for the reaction.
- Is the reaction thermodynamically favorable at 298 K? Justify your answer based on the calculation of the value and sign of ΔG° .
- Is the value of the equilibrium constant, K , for the reaction greater than 1, or less than 1? Justify your answer.

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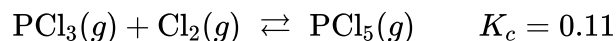
70. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to PCl_3 .

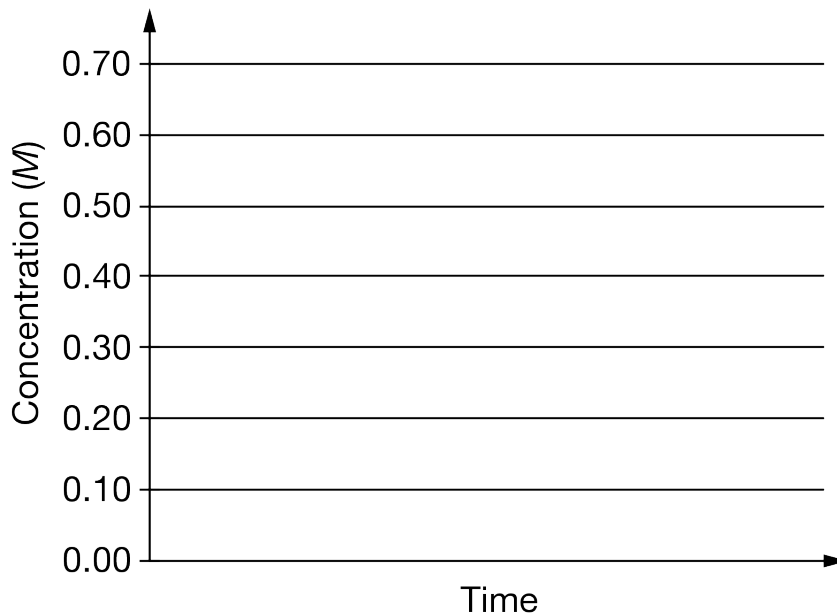
- (a) In the box below, draw the complete Lewis electron-dot diagram for PCl_3 .



- (b) What is the shape of the PCl_3 molecule?



A 0.60 mol sample of $\text{PCl}_3(g)$ and a 0.70 mol sample of $\text{Cl}_2(g)$ are placed in a previously evacuated 1.0 L rigid container, and the reaction represented above takes place. At equilibrium, the concentration of $\text{PCl}_5(g)$ in the container is 0.040 M.



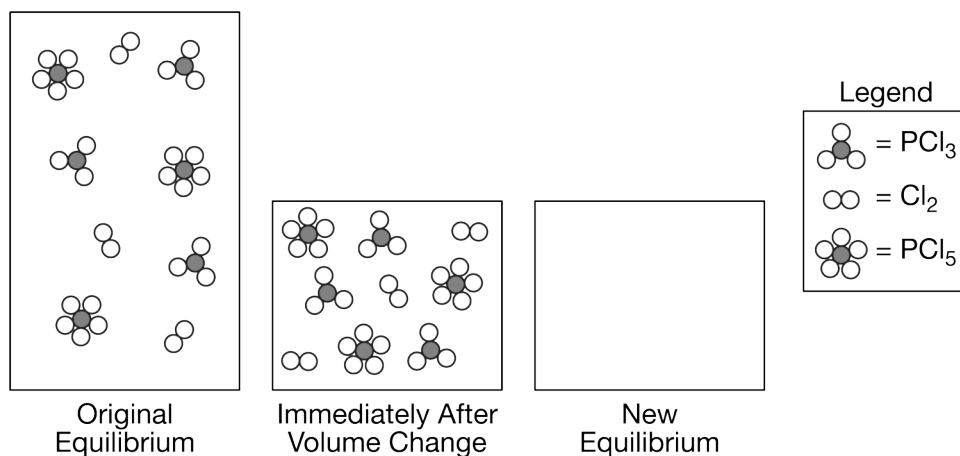
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(c) On the graph above, carefully draw three curves, one for each of the three gases. The curves must show how the concentration of each of the three gases changes as equilibrium is established. Label each curve with the formula of the gas.

(d) Explain how the slopes of the curves at equilibrium can be used to show that the rates of the forward and reverse reactions are the same at equilibrium.

(e) As the reaction occurs at constant temperature, does the pressure inside the container increase, decrease, or remain the same? Explain.

Another mixture of $\text{PCl}_3(g)$, $\text{Cl}_2(g)$, and $\text{PCl}_5(g)$ is at equilibrium at a different temperature in a different vessel. The system at equilibrium is represented by the following box on the left. Then the volume of the vessel is decreased. The box in the middle represents the system before equilibrium is reestablished at the same temperature.



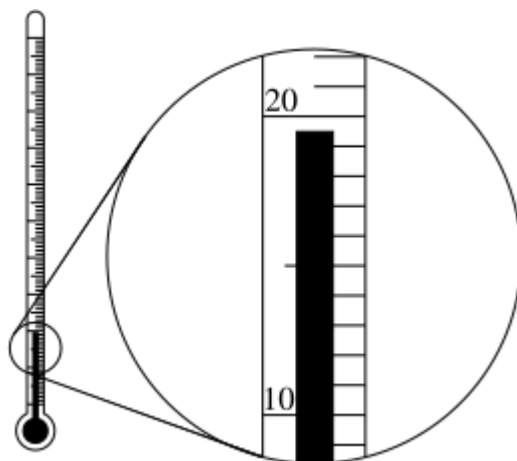
(f) In the box above on the right, draw an appropriate number of each type of molecule to represent a possible new equilibrium.

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71. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

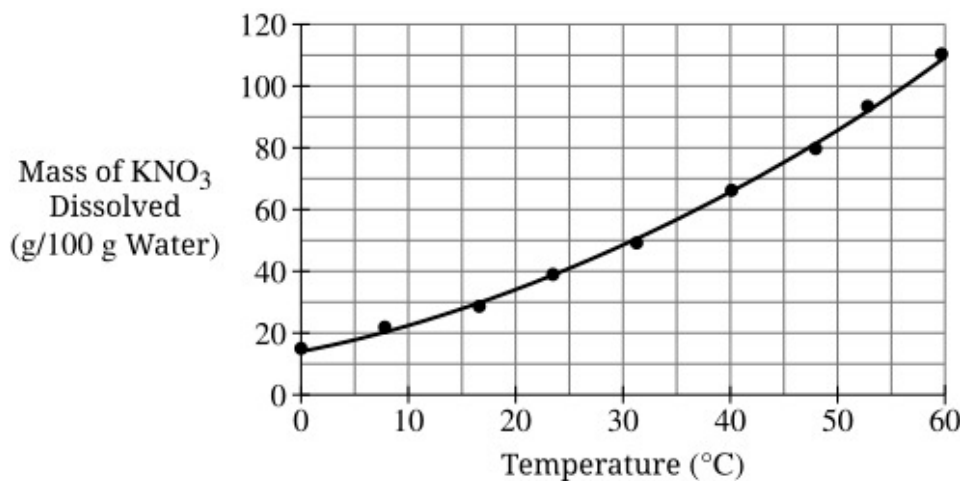
A student conducts a solubility experiment with potassium nitrate, KNO_3 , in water by identifying the temperature at which various masses of KNO_3 will completely dissolve.

(a) At room temperature, H_2O is a liquid and KNO_3 is a solid. The student incorrectly claims that this is because potassium and nitrate ions experience stronger dispersion forces than H_2O molecules. In terms of the types of attractive forces, explain why the student is incorrect.



(b) In one of the trials, the student reads the initial temperature of the water using the Celsius thermometer above. Record this temperature to the appropriate number of significant figures.

(c) At the temperature indicated on the thermometer, the student adds the solid KNO_3 to the sample of water and begins to stir the mixture. Explain the effect of stirring on the maximum amount of solid that can dissolve in the water.



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The graph above summarizes the data collected by the student.

(d) Another student places 38.1 grams of KNO_3 into 100.0 g of water at the temperature measured in part (b), stirs the mixture, and observes that the solid does not dissolve completely. The student plans to heat the mixture and states that the solid will completely dissolve before the mixture reaches 30°C . Do you agree with the claim? Justify your answer with information from the graph.